

neoplasm was also included. Histopathologic analysis revealed chronic inflammation and polypoid tissue with viable reactive bone. After endoscopy and histopathologic evaluation, the patient was prescribed 300 mg chlorhexidine and sodium chloride solution.

DISCUSSION/CONCLUSIONS: There are reports of sinus involvement with MRONJ if the radiograph shows unilateral mucoperiosteal thickening, sinus opacification, and fistula formation. These findings may mimic sinusitis. The patient had these radiographic findings as well as discontinuity of the sinus boundaries.

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CONCURRENT METASTATIC DISEASE AND MEDICATION RELATED OSTEONECROSIS OF THE JAWS

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BACKGROUND: Antiresorptive medications are often used in the management of cancer and cancer-related conditions including primary and metastatic disease. A rare but well-established complication of antiresorptives is osteonecrosis of the jaws (ONJ). ONJ can manifest with radiographic features similar to those of metastatic diseases, which makes differentiating between these diseases a diagnostic challenge.

CASE REPORT: We present the pertinent radiologic features from 11 patients with concurrent metastatic diseases and ONJ. The radiologic appearances of the metastatic lesions varied from well-defined radiolucencies to radiopacities with periosteal bone deposition. The areas affected by ONJ shared a common spectrum of osseous changes, demonstrating lytic defects, sequestration of bone, diffuse sclerosis, and periosteal bone formation.

DISCUSSION/CONCLUSIONS: Metastatic disease and ONJ possess similar radiographic features, with patchy lytic and sclerotic changes in the affected region of bone. Likewise, both entities evoke periosteal reactions, although the pattern may vary. Although exposed bone in the oral cavity for 8 weeks or longer was once considered the hallmark for diagnosis of MRONJ, the American Association of Oral and Maxillofacial Surgeons modified the definition in 2014 to include a Stage 0, where radiographic changes are apparent without any bone exposure. Similarly, the absence of specific radiologic appearances

complicates the differentiation between these 2 entities in the establishment of MRONJ.

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DETECTING APICAL RADIOLUCENCIES USING DEEP LEARNING TECHNOLOGY: A PILOT STUDY

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BACKGROUND: Dentists rely on radiographs in making treatment decisions every day. However, if inattentively read, radiographs can lead to over- or undertreatment.

OBJECTIVE(S): The objectives were to assess the diagnostic performance of a deep learning algorithm at detecting apical radiolucencies (ARs) on intraoral (IO) radiographs and compare the performance of the algorithm vs interpretation by experts.

STUDY DESIGN: A search was conducted using the University of North Carolina cone beam computed tomography (CBCT) referral database for volumes acquired for endodontic evaluation. The inclusion criteria consisted of permanent teeth that exhibited apical radiolucencies measuring ≥ 2 mm on CBCT with a diagnostic IO radiograph for the same site. The exclusion criteria included patients under 18 years old and CBCT and IO images acquired more than 6 months apart. After applying the inclusion and exclusion criteria, 192 cases were found. The CBCT volumes served as ground truth for the IO radiograph annotation. The IO radiographs were randomly assigned to one of 3 groups: 6 images, observer calibration; 54 images, training the artificial intelligence (AI) algorithm; 132 images, testing set. Additionally, 132 IO radiographs with no evidence of apical radiolucencies were included to serve as controls. For this pilot, 70 images were randomly selected to create a testing subset derived from the main control and positive testing sets. Three experts were presented with this testing subset and asked to independently determine locations of AR and confidence of presence using a 5-point Likert scale.

RESULTS: The standalone software performance results by tooth were as follows: Sensitivity was 93%, specificity was 88%, and the area under the receiver operating characteristic curve (ROC-AUC) was 94% (95% confidence interval [CI], 89%, 98%). The combined results of the experts were as follows: sensitivity was 87%, specificity was 97%, and ROC-AUC was 93% (95% CI, 88%, 98%). Notably, the software performance metrics were lowest in the maxillary posterior region.

DISCUSSION/CONCLUSIONS: Using a limited testing data set, AI provided comparable performance to expert observers for this task. Further AI training is necessary to increase the sensitivity and specificity of AR detection in the posterior maxillary region.